Exploratory Data Analysis: Variation

MKT 566 · Fall 2026

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What is EDA?

From charts to questions

- Last week you built the **vocabulary**: which chart types exist, what each is for, and how to make a figure worth showing
- This week you put them to work: using charts **systematically** to understand a dataset you have never seen before
- That process has a name: **Exploratory Data Analysis** (EDA)

Partially based on Chapter 7 of R for Data Science (this week's required reading).

What we will learn

How to use visualization to explore your data in a systematic way:

1. **Generate questions** about your data
2. **Search for answers** by visualizing, transforming, and modeling your data
3. Use what you learn to **refine** your questions and/or generate new ones

EDA goal

- Develop an **understanding of your data**
- The easiest way to do this is to use **questions as tools** to guide your investigation
- EDA is fundamentally a **creative process**

> There is no fixed recipe: the quality of your EDA depends on the quality of the questions you ask, and each answer suggests the next question.

Start with summary statistics

Computing summary statistics is also very useful and should be done at the beginning of any analysis:

- Mean, median
- Standard deviation
- Min, max
- Number of **missing values**

These descriptive statistics can provide valuable insights into your data, and they are one line of R (`summary()`, as we will see shortly).

Data cleaning

During EDA you will perform what is called **data cleaning**, which involves things like:

- Finding and removing **erroneous data** (impossible ages, negative prices, duplicated rows)
- Deciding what to do with **outliers**
- Deciding what to do with **missing data**

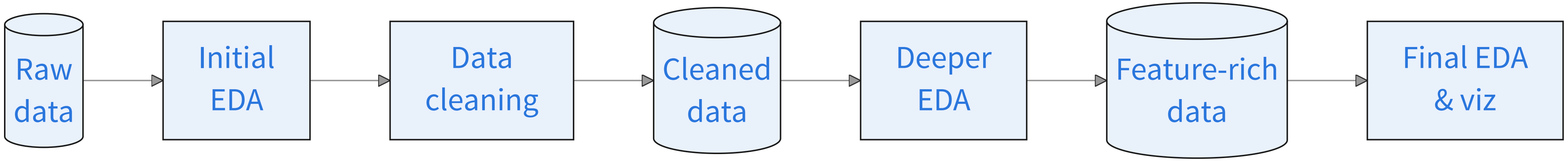
Rule of thumb: you will spend far more time cleaning than modeling. Budget for it.

Feature engineering

Very often you will create new variables from existing ones. This process is referred to as **feature engineering**, e.g.:

- Extract the **week number** from a date variable
- Sum up **total ad spend** across all advertising channels
- Compute the **cumulative average review rating** for each product

The EDA workflow



- **Initial EDA**: spot errors, outliers, and missing data → clean them up
- **Deeper EDA**: uncover patterns and group effects, engineer features

The loop matters: what you find in deeper EDA often sends you back to cleaning.

The rules of EDA

Elea McDonnell Feit · 1st
Professor of Marketing and Associate Dean of Research
3w • Edited •

Rule 1: Never trust your data. There are all sorts of oddities in any data set and you want to learn about them.

Rule 2: Summarize and plot your data. Over and over. All kinds of summaries. Not just the default summaries in the software you are using. Ask questions like, "How many subscriptions are there that last more than 10 years?" and "How prominent is the holiday seasonality?"

Rule 3: Compare the summaries to your priors. When something in the data seems seems off, pause and consider the consequences for your analysis. Keep a list of these oddities in your lab notebook.

Rule 4: Chase oddities that might be consequential. Form a hypothesis that explains what is going on. Then think of a way to test it. Do more summaries. Look at the documentation again. Ask someone. Observe the process the data describes. Don't stop until you are pretty sure you know what is going on.

R scripts for this week

In the `code/` folder on the course website:

1. **`w2-1-eda-variation-class.R`**: reproduces every chart in today's deck
2. **`w2-1-simulate-marketing-dataset.R`**: generates the case dataset, `data/marketing_eda.csv` (you do not need to run it)

> Download everything as one zip: **`w2-code.zip`**, open the scripts in VS Code, and run along.

The in-class case is a separate handout, and it has **no starter code** on purpose: you will vibecode it. More on the last slides.

How Do We Explore the Data?

Variation and covariation

- There is **no rule** about which questions you should ask to guide your exploration
- However, two types of questions will always be useful for making discoveries within your data:

> 1. What type of **variation** occurs *within* my variables? *(today)*
> 2. What type of **covariation** occurs *between* my variables? *(next week)*

What is variation?

Variation is the tendency of the values of a variable to change from measurement to measurement.

- We are interested in "**within-the-same-variable**" patterns

We can observe variation:

- For the same **continuous** variable over two different measures, e.g., temperature measured at 10 am and 4 pm
- For the same **categorical** variable across different "subjects", e.g., eye color across individuals

How to visualize variation

The best way to understand patterns of variation is to visualize:

1. The **distribution** of the variable's values
2. How the variable **evolves over repeated observations** (e.g., when we have repeated observations over time, i.e., panel data or time series)

We will look at both, using two datasets.

Visualizing distributions

Different charts depending on the type of variable:

Variable type	Chart
Continuous (sales, ad spend, prices)	Histogram, density plot
Categorical / discrete (channel, device, star rating)	Bar chart

And for spotting outliers: the **box plot** (coming up).

Visualizing Distributions

The `marketing` dataset

From the R library **`datarium`**: a marketing experiment with the advertising budget spent on three channels and the resulting sales.

```r
library(datarium)
head(marketing)
```

```
  youtube facebook newspaper sales
1  276.12    45.36     83.04 26.52
2   53.40    47.16     54.12 12.48
3   20.64    55.08     83.16 11.16
4  181.80    49.56     70.20 22.20
5  216.96    12.96     70.08 15.48
6   10.44    58.68     90.00  8.64
```

```r
nrow(marketing)
```

```
[1] 200
```

200 observations. Budgets (`youtube`, `facebook`, `newspaper`) and `sales` are in thousands of dollars / units.

Summary statistics first

```r
summary(marketing)
```

```
    youtube          facebook       newspaper          sales      
 Min.   :  0.84   Min.   : 0.00   Min.   :  0.36   Min.   : 1.92  
 1st Qu.: 89.25   1st Qu.:11.97   1st Qu.: 15.30   1st Qu.:12.45  
 Median :179.70   Median :27.48   Median : 30.90   Median :15.48  
 Mean   :176.45   Mean   :27.92   Mean   : 36.66   Mean   :16.83  
 3rd Qu.:262.59   3rd Qu.:43.83   3rd Qu.: 54.12   3rd Qu.:20.88  
 Max.   :355.68   Max.   :59.52   Max.   :136.80   Max.   :32.40  
```

One line of R. What do you notice?

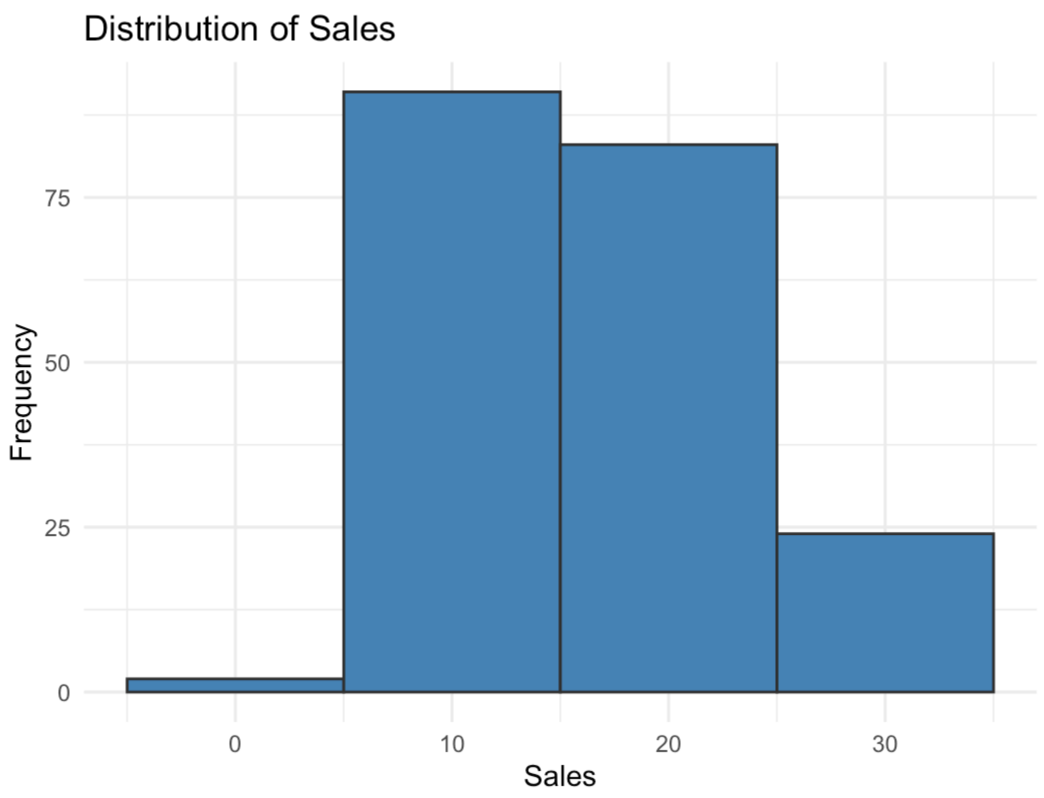
Summary statistics: what they tell us

```
    youtube          facebook         newspaper          sales      
 Min.   :  0.84   Min.   : 0.00   Min.   :  0.36   Min.   : 1.92  
 1st Qu.: 89.25   1st Qu.:11.97   1st Qu.: 15.30   1st Qu.:12.45  
 Median :179.70   Median :27.48   Median : 30.90   Median :15.48  
 Mean   :176.45   Mean   :27.92   Mean   : 36.66   Mean   :16.83  
 3rd Qu.:262.59   3rd Qu.:43.83   3rd Qu.: 54.12   3rd Qu.:20.88  
 Max.   :355.68   Max.   :59.52   Max.   :136.80   Max.   :32.40  
```

- **No missing values** in any column
- `facebook` spend starts at **zero**: some campaigns skip it entirely
- `youtube` gets by far the **largest budgets**
- `sales` vary by a factor of **~17** across campaigns

Distribution of sales

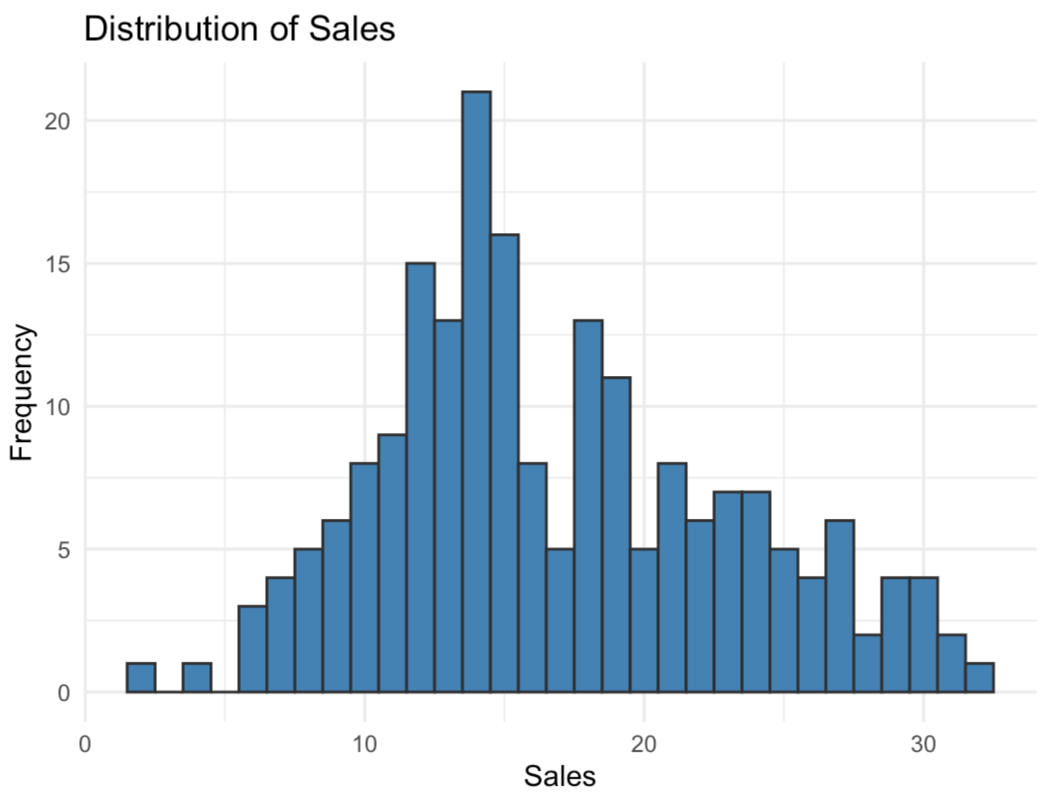
```r
ggplot(marketing, aes(x = sales)) +
  geom_histogram(
    binwidth = 10,
    fill = "steelblue", color = "grey20"
  ) +
  labs(title = "Distribution of Sales",
       x = "Sales", y = "Frequency") +
  theme_minimal()
```



`binwidth` controls how wide each bar is: here each bar covers 10 units of sales.

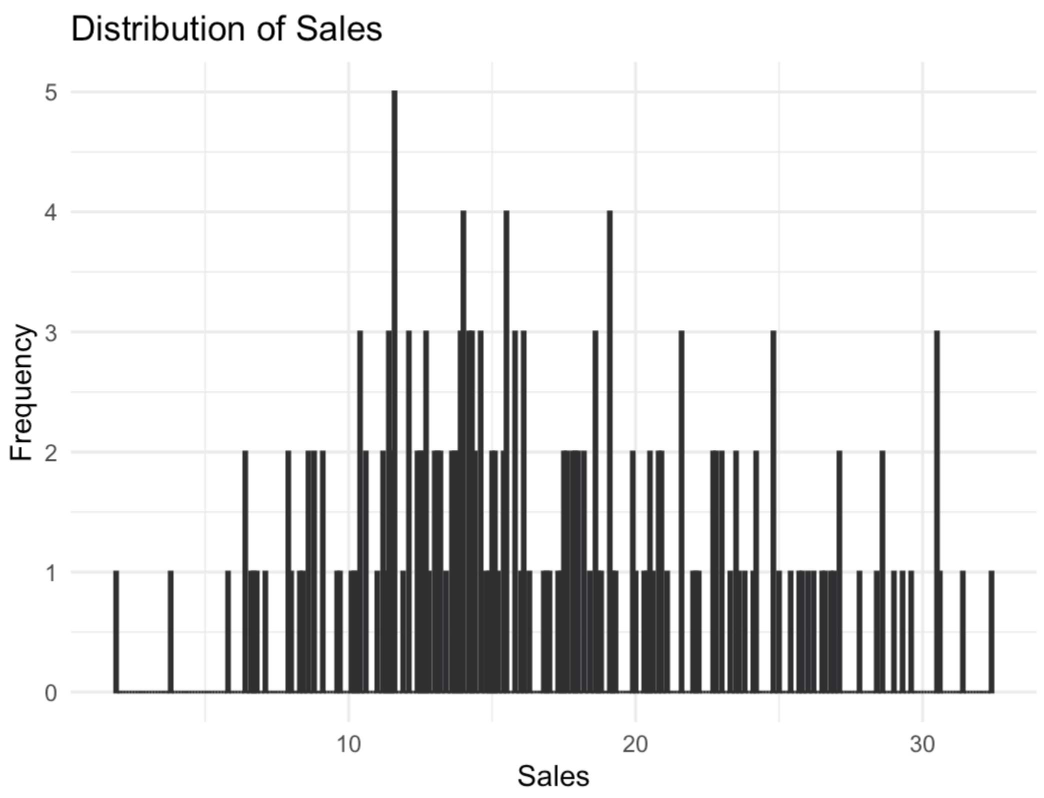
Distribution of sales: smaller bins

```r
ggplot(marketing, aes(x = sales)) +
  geom_histogram(
    binwidth = 1,
    fill = "steelblue", color = "grey20"
  ) +
  labs(title = "Distribution of Sales",
       x = "Sales", y = "Frequency") +
  theme_minimal()
```



Distribution of sales: too small

```r
ggplot(marketing, aes(x = sales)) +
  geom_histogram(
    binwidth = 0.1,
    fill = "steelblue", color = "grey20"
  ) +
  labs(title = "Distribution of Sales",
       x = "Sales", y = "Frequency") +
  theme_minimal()
```

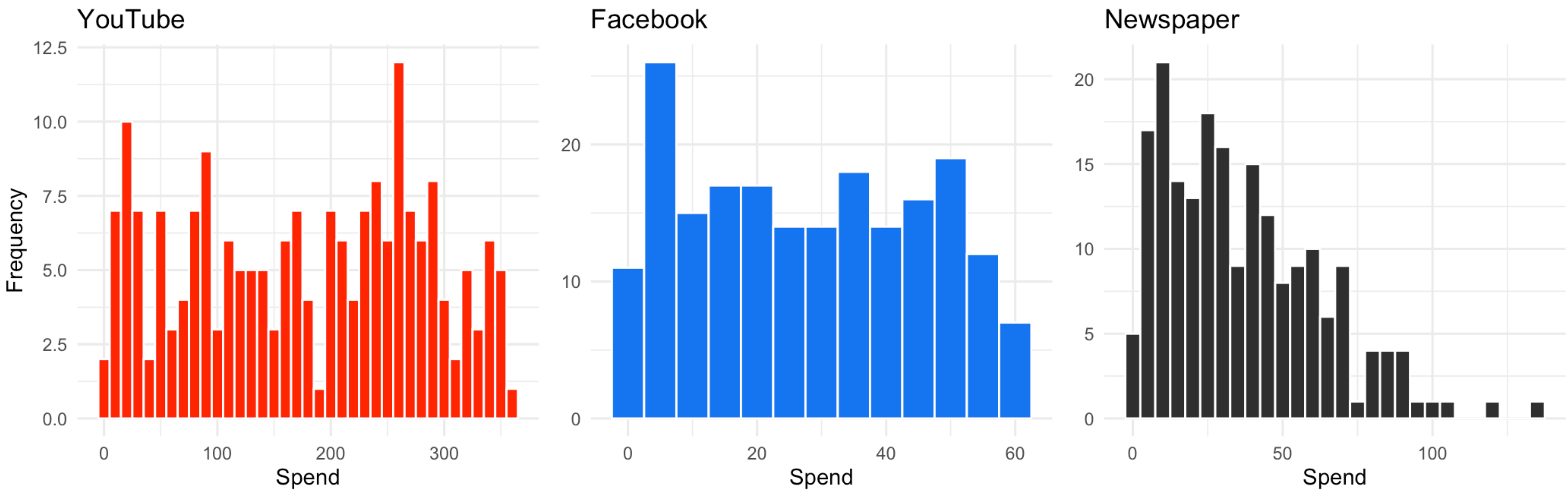


Binwidth: what did we just see?

- `binwidth = 10`: too coarse. Three bars, the shape is hidden
- `binwidth = 1`: the shape appears: most weeks sell 10–20, with a right tail of strong weeks
- `binwidth = 0.1`: too fine. Mostly noise

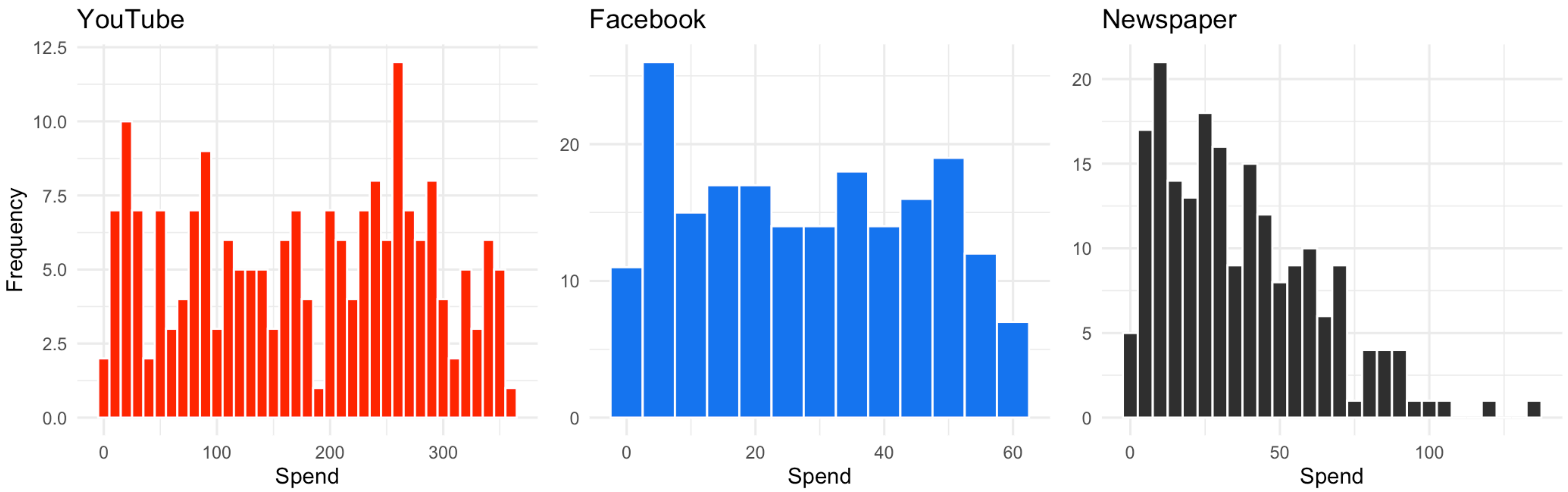
> Bin width is a **choice**, and it changes what you see: too wide hides the shape, too narrow shows noise. Always try a few values.

Spend across the three channels



Same variable (ad spend), three channels. What do you see?

Spend across the three channels: three shapes



- **YouTube**: spread almost uniformly across its range
- **Facebook**: flat-ish, with a mode at low spend
- **Newspaper**: **right-skewed**, a few campaigns spend a lot

What can we learn from distribution charts?

- **Common vs. rare values**: high vs. low bars
- **Shape**: symmetric or skewed? One mode or several?
- **Unusual patterns**: gaps, spikes, values that should not exist
- **Candidate outliers**: bars far from the rest

The last one deserves its own tool…

Outliers

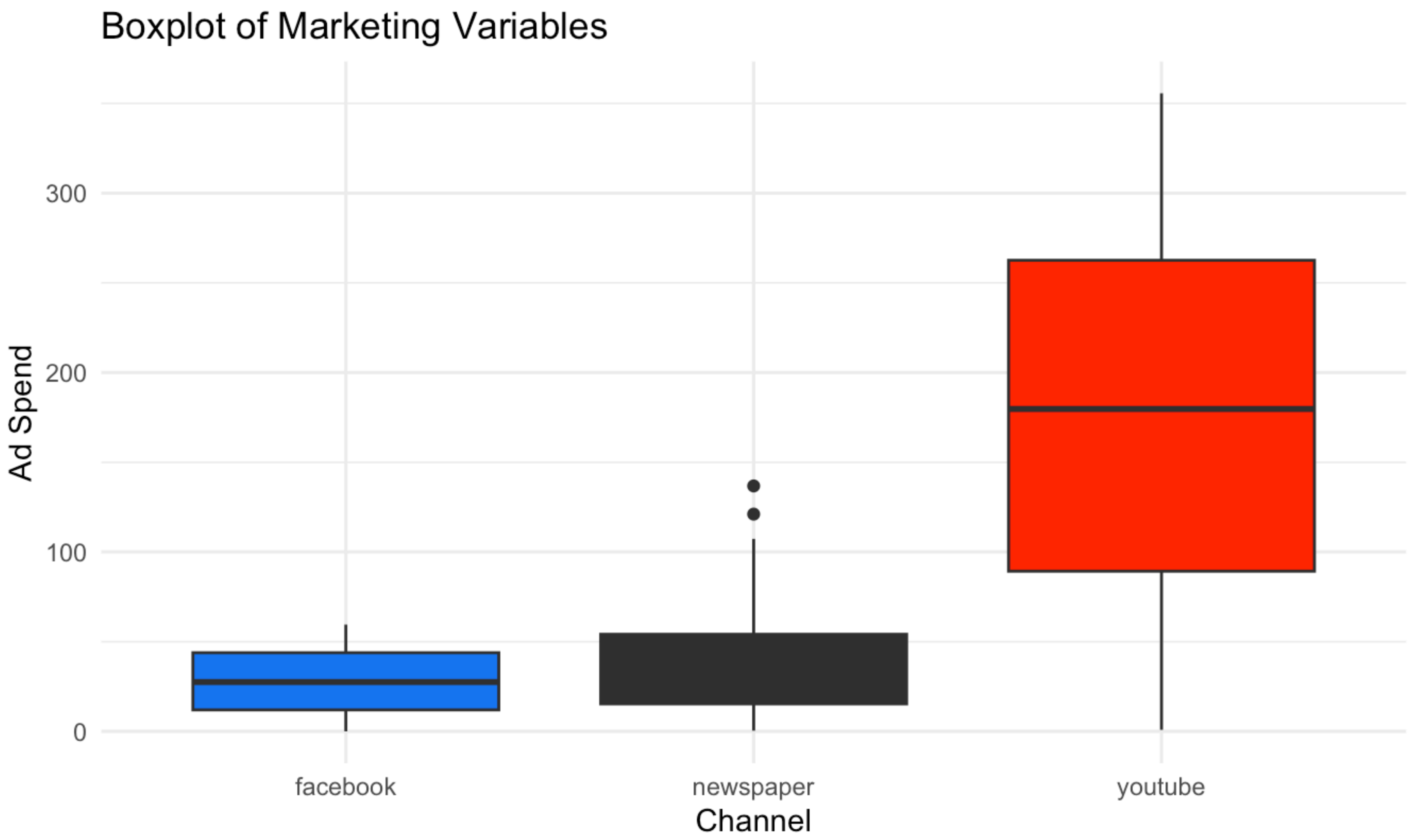
How to specifically look for outliers

Box plots (or scatter plots) are often more useful than histograms:

- Box plots automatically flag outliers using a mathematical rule
- Any point beyond **1.5 × IQR** (interquartile range):
  - IQR = Quartile 3 − Quartile 1
  - Lower bound = Q1 − 1.5 × IQR
  - Upper bound = Q3 + 1.5 × IQR

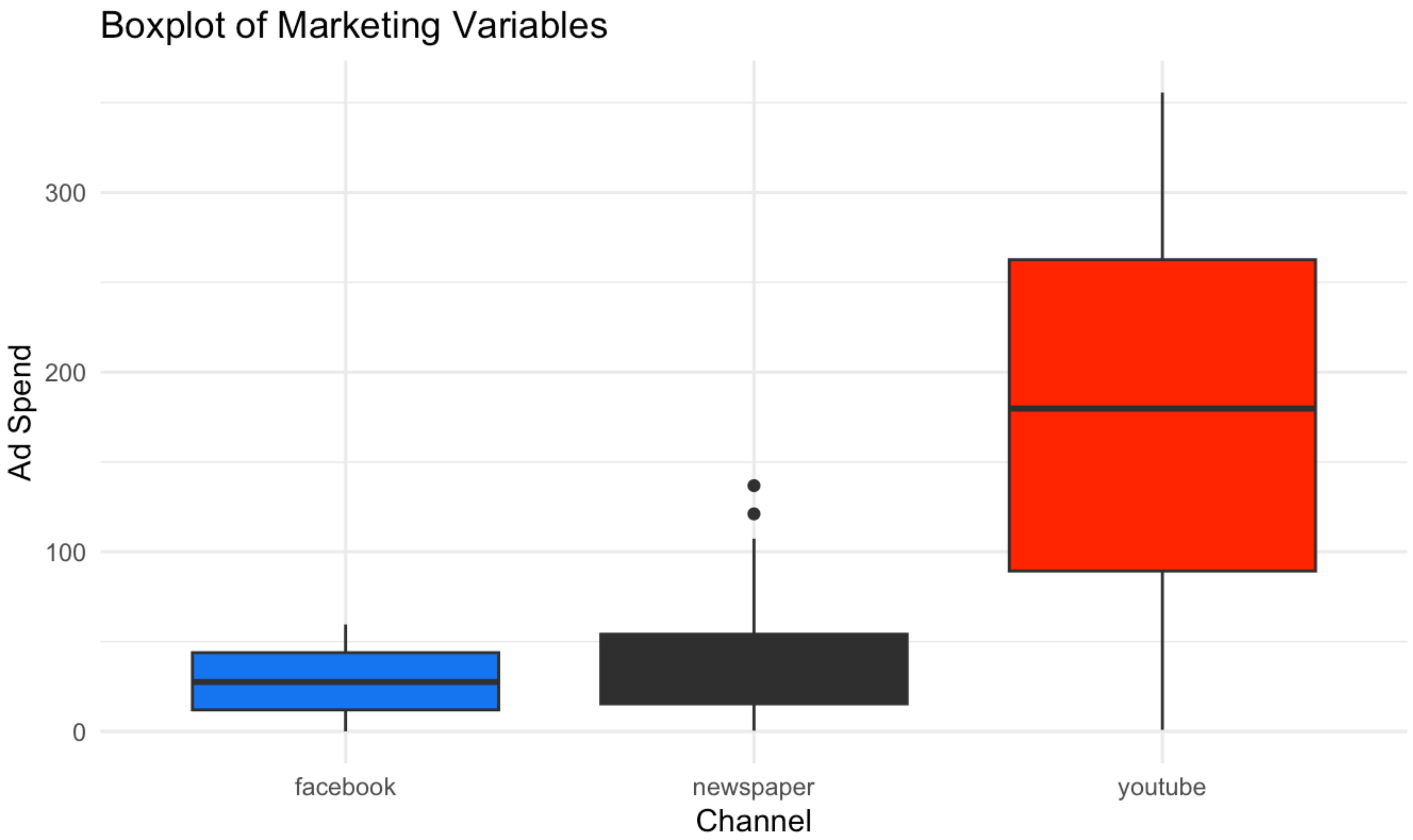
Recall from last week: the box is the middle 50% of the data, the line is the median, the whiskers reach the last point inside the bounds, and dots beyond them are flagged as outliers.

Example with ad spend



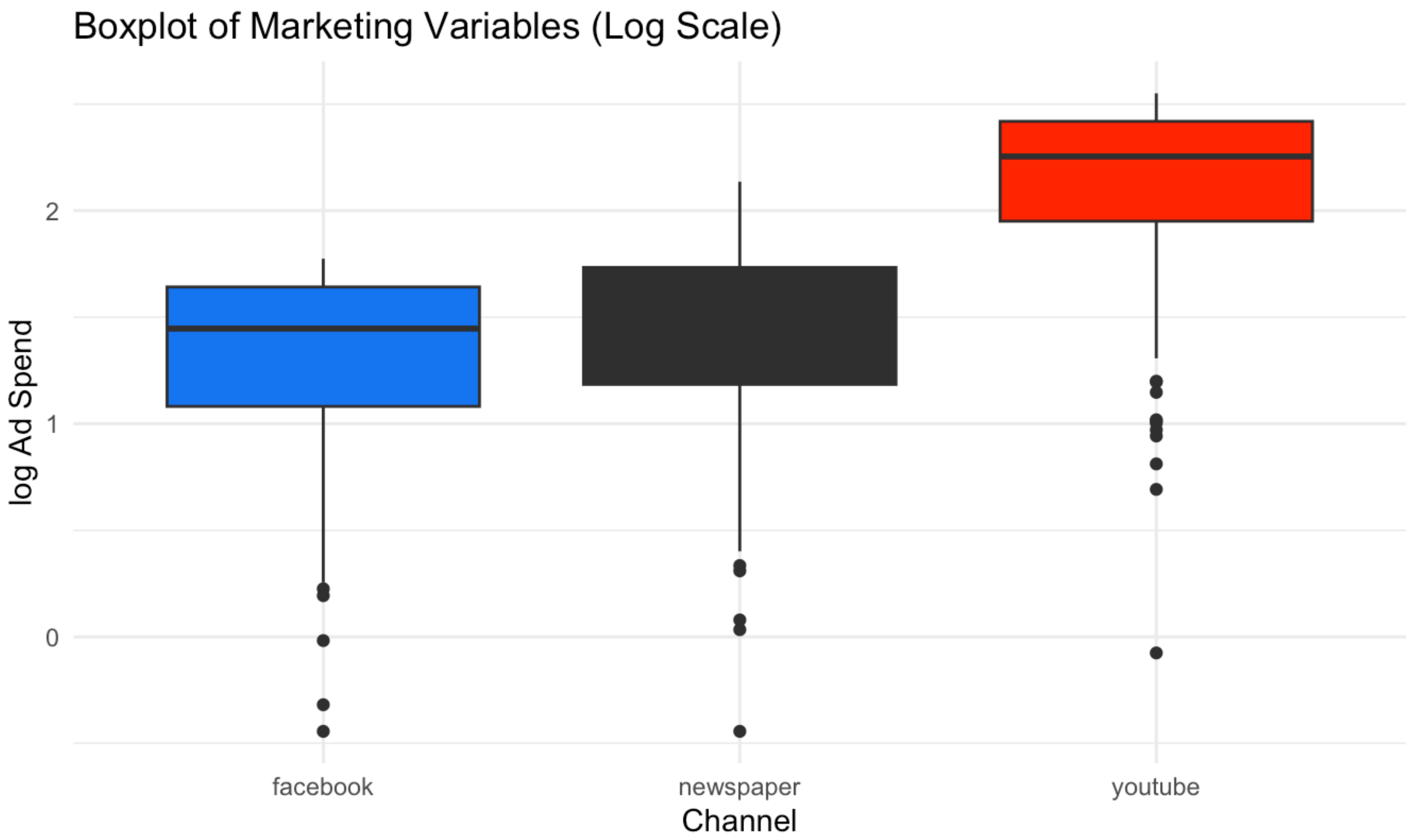
What do you see?

Example with ad spend: reading it



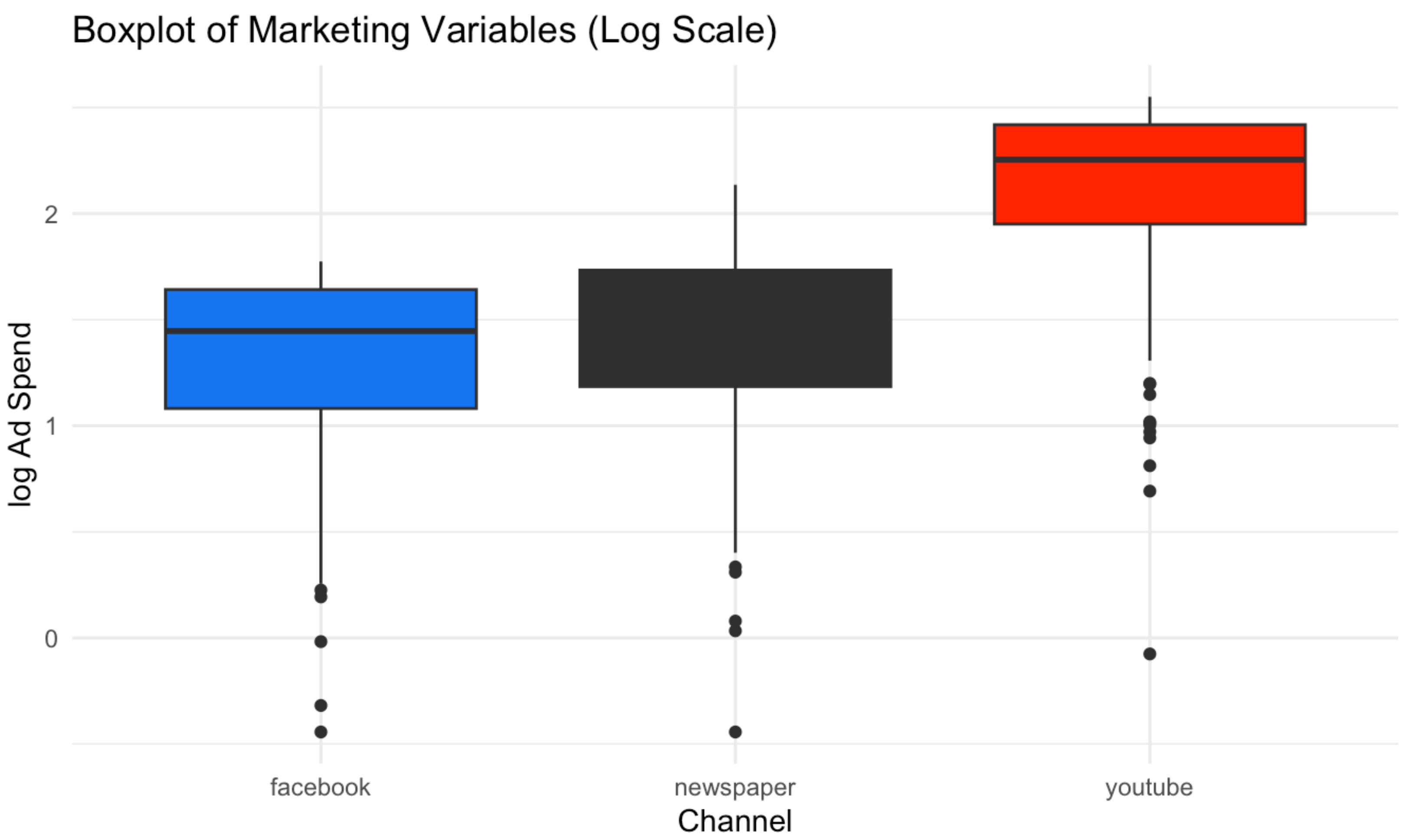
- **Newspaper** shows flagged outliers on the high side
- But the three channels live on very **different scales**, so the smaller ones are squashed against the axis

Example with logged ad spend



Same data, log scale. What changed?

Example with logged ad spend: reading it



- On the log scale the comparison across channels is **fair**
- A different picture appears: now **low-spend** outliers become visible too

What the log transformation does

Log makes small values more visible: it expands small values and compresses large ones.

When and why to use log:

- To handle **skewed data** (common in marketing: spend, sales, and clicks distributions)
- To make patterns at the **low end** visible when a few big values dominate
- To turn **multiplicative growth** into a straight line (e.g., exponential trends)

Why logs matter in EDA

- Log makes small values visible: you don't lose them at the bottom of the plot
- It can also reveal **negative outliers** (very low spend, clicks, sales) that were hidden on the linear scale
- It emphasizes **relative differences (ratios)** instead of absolute differences: $\log(a) - \log(b) = \log(a/b)$

Example: sales increase from \$100 to \$110

- Actual % change: $\frac{110-100}{100} = 0.10 \rightarrow$ **10%**
- Using logs: $\log(110) - \log(100) = \log(1.1) \approx 0.095 \rightarrow$ **9.5%**

For small changes, a log difference **is** (approximately) the % change. We will use this all the time when we get to regression.

Best practice for outliers

- It's good practice to **repeat your analysis with and without** the outliers
- If they have **minimal effect** on the results, and you can't figure out why they're there, it's reasonable to replace them with missing values and move on
- If they have a **substantial effect**, you shouldn't drop them without justification: figure out what caused them (e.g., a data entry error) and **disclose** that you removed them in your write-up
- Generally, rely on **robust statistics** (median over mean, quantiles over ranges)

Visualizing Trends

The `economics` dataset

The second way to visualize variation: how a variable **evolves over time**. Example with ggplot2's `economics` dataset, monthly US indicators from FRED:

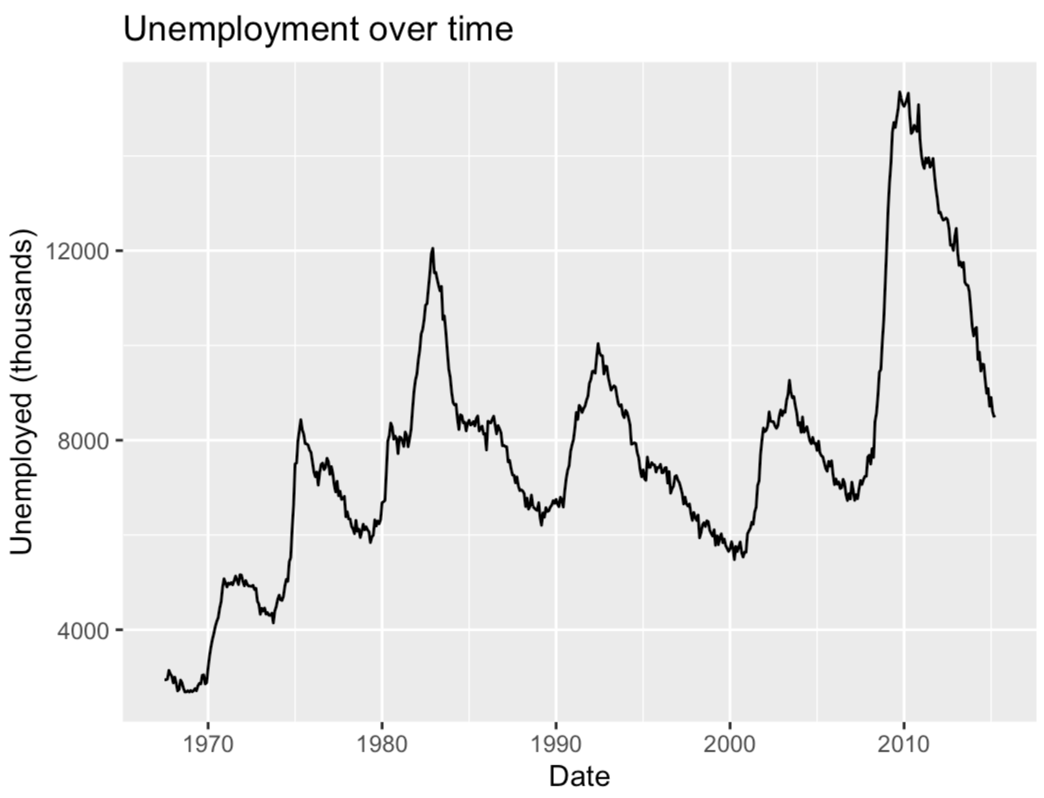
```r
head(economics)
```

```
# A tibble: 6 × 6
  date         pce    pop psavert uempmed unemploy
  <date>     <dbl>  <dbl>   <dbl>   <dbl>    <dbl>
1 1967-07-01  507. 198712    12.6     4.5     2944
2 1967-08-01  510. 198911    12.6     4.7     2945
3 1967-09-01  516. 199113    11.9     4.6     2958
4 1967-10-01  512. 199311    12.9     4.9     3143
5 1967-11-01  517. 199498    12.8     4.7     3066
6 1967-12-01  525. 199657    11.8     4.8     3018
```

`unemploy` is the number of unemployed, in thousands.

Unemployment over time

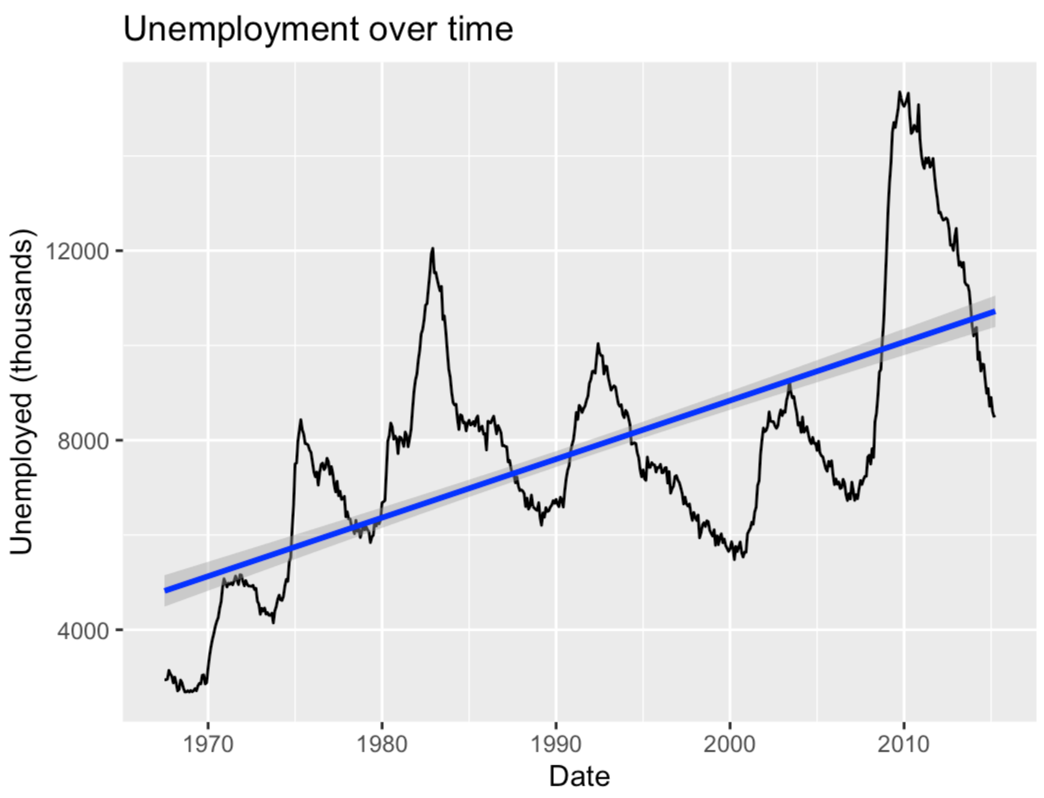
```r
ggplot(data = economics) +
  geom_line(
    mapping = aes(x = date, y = unemploy)
  ) +
  labs(title = "Unemployment over time",
       x = "Date",
       y = "Unemployed (thousands)")
```



A line chart is the default for time series. What patterns do you see?

Long-run direction

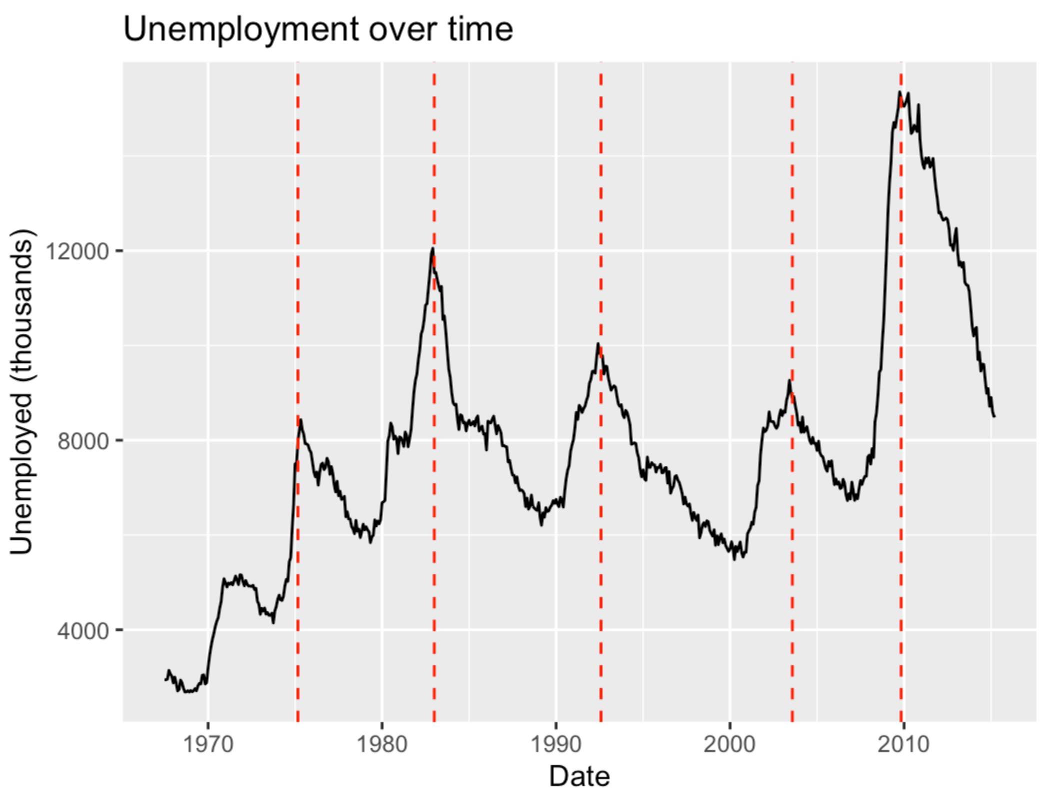
```r
ggplot(data = economics) +
  geom_line(
    mapping = aes(x = date, y = unemploy)
  ) +
  labs(title = "Unemployment over time",
       x = "Date",
       y = "Unemployed (thousands)") +
  geom_smooth(
    mapping = aes(x = date, y = unemploy),
    method = "lm", color = "blue"
  )
```



A fitted line summarizes the **long-run directional trend**: unemployment drifts up as the labor force grows.

Cycles and anomalies

```r
ggplot(data = economics) +
  geom_line(
    mapping = aes(x = date, y = unemploy)
  ) +
  labs(title = "Unemployment over time",
       x = "Date",
       y = "Unemployed (thousands)") +
  geom_vline(
    xintercept = as.Date(c(
      "1975-03-01", "1983-01-01", "1992-08-01"
      "2003-08-01", "2009-11-01")),
    color = "red", linetype = "dashed"
  )
```



Annotating events (here, unemployment peaks after recessions) turns a line into a story: each spike has a cause.

What can we learn from trends?

- **Long-run directional trends**: is the series drifting up, down, or flat?
- **Seasonality & cyclicality**: regular patterns that repeat (weekly, yearly, business cycles)
- **Anomalies & outliers**: spikes and drops that demand an explanation, often the most interesting finding in the data

> In marketing: sales trends, seasonality in demand, and campaign-driven spikes are exactly these three patterns. You will look for them in every dataset you touch.

The Variation Case

The variation case

In class you will apply all of this to a fresh dataset: **1,000 customers** of an online retailer, with age, gender, device, ad channel, ad spend, clicks, purchases, and revenue.

Here is the twist: **there is no starter code**. You will **vibecode** the whole analysis, exactly as we discussed: you describe the chart, your AI assistant writes the R, you run it, look, refine, and **you** interpret.

You will answer questions like:

- Is ad spend **symmetric or skewed**, and what marketing strategy could create that shape?
- Is the data **balanced across channels**?
- Are there **outliers**, and should we drop them?

How the case works

The handout (`w2-1-variation-case.html`) takes the training wheels off gradually:

1. **Tasks 1–2** (look at the data, summary stats): we give you the exact prompts to paste
2. **Tasks 3–4**: a prompt skeleton with blanks, then just a hint
3. **Tasks 5–7**: only the goal; the prompt is yours (the last task is a question **you** invent)

Deliverables, both made with your assistant: an **R Markdown report knitted to HTML** with your charts and your written answers (the exact format you will submit for the homeworks), and your `ai-log.md`.

Wrap-up

Code and data to reproduce today's charts, all on the course website:

- `w2-code.zip`: one-click download of everything below
- `w2-1-eda-variation-class.R`: every chart shown today
- `w2-1-simulate-marketing-dataset.R`: generates the case dataset
- `data/marketing_eda.csv`: the case dataset
- **The variation case handout**: the in-class vibecoding exercise (bring your laptop!)

Required reading: Chapter 7 of R for Data Science. Optional: Chapters 3–5 of *R for Marketing Research and Analytics*.

Questions?